

Tianyue Zhou

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EDUCATION

Massachusetts Institute of Technology (MIT) <i>Ph.D. in Civil and Environmental Engineering; Advisor: Prof. Cathy Wu; GPA: 5.0/5.0</i>	Sept 2025 - Present Cambridge, MA
ShanghaiTech University <i>B.Eng. in Computer Science; GPA: 3.75/4.0</i>	Sept 2021 - June 2025 Shanghai, China
Massachusetts Institute of Technology (MIT) <i>Undergraduate Exchange Student, Computer Science; GPA: 4.5/5.0</i>	Sept 2023 - Dec 2023 Cambridge, MA

RESEARCH INTERESTS

- **Machine Learning:** Reinforcement Learning, Multi-task Learning, Transfer Learning.

PUBLICATIONS

- **Expert with Clustering: Hierarchical Online Preference Learning Framework** [paper]
Tianyue Zhou, Jung-Hoon Cho, Babak Rahimi Ardabili, Hamed Tabkhi, Cathy Wu
Learning for Dynamics and Control Conference, 707–718
- **The Nah Bandit: Modeling User Non-compliance in Recommendation Systems** [paper] [website] [code]
Tianyue Zhou, Jung-Hoon Cho, Cathy Wu
IEEE Transactions on Control of Network Systems (Volume: 12, Issue: 4, Dec. 2025), 2919-2931
- **Structure Detection for Contextual Reinforcement Learning** [paper] [website] [code]
Tianyue Zhou, Jung-Hoon Cho, Cathy Wu
Proceedings of the AAAI Conference on Artificial Intelligence, 40(34), 29009–29016.

RESEARCH EXPERIENCE

Massachusetts Institute of Technology <i>Undergraduate Research Assistant, Supervised by Prof. Cathy Wu, MIT LIDS</i>	Cambridge, MA
• Online Preference Learning for Drivers – Objective: Implement online learning algorithm in a travel route recommendation problem, aiming to rapidly capture user preferences and recommend personalized and eco-friendly travel routes. – Developed a novel contextual bandit framework, Expert with Clustering (EWC), leveraging hierarchical user information for rapid and accurate learning of user preferences, achieving a 27.57% performance improvement over LinUCB baseline in experiments. – Proposed a distance metric, Loss-guided Distance, tailored for the online preference learning problem, which enhances the representativeness of centroids in clustering. – Established the regret bound of EWC, indicating superior theoretical performance compared to LinUCB. • Modeling User Non-compliance in Recommendation Systems – Objective: Tackle a critical challenge often overlooked by traditional frameworks in physical-world recommendation problems: users can easily dismiss recommendations that do not appeal to them and revert to their baseline behavior. – Introduced a novel bandit problem–Nah Bandit–for modeling user non-compliance in recommendation systems. This framework offers the potential to accelerate preference learning compared to traditional frameworks. – Proposed a user non-compliance model to parameterize the anchoring effect in Nah Bandit, reducing the bias when learning user preference. – Combined the user non-compliance model with EWC, further efficiently utilizing non-compliance feedback and hierarchical information. Experimental results highlight that EWC achieves at least 15.36% performance improvement in multiple applications compared to both supervised learning and traditional bandits. • Structure Detection for Contextual Reinforcement Learning – Objective: Extend model-based transfer learning approach to address multi-dimensional contextual reinforcement learning (cRL) problems. – Proposed SD-MBTL, a generic framework that detects underlying generalization structures and adapts task selection strategies accordingly.	Sept 2023 - Jan 2024 Jan 2024 - June 2025 Aug 2024 - Aug 2025

- Developed M/GP-MBTL, an initialization of SD-MBTL that combines a clustering based approach and a Gaussian Process-based approach, tailored to different structural patterns in CMDPs.
- Validated on multi-dimensional real-world benchmarks, including robotics, eco-driving, and agriculture, showing significant gains in generalization and sample efficiency.

HONORS AND AWARDS

- ShanghaiTech **Outstanding Student** (2021-2022), Top **3%** ~ **7%** *Dec 2022*
- ShanghaiTech **Outstanding Student Union Officer** (2021-2022) *Dec 2022*
- **Scholarship** for Undergraduate 3+1 Overseas Exchange Program, ~\$9,000 *June 2024*

MISCELLANEOUS

- **Programming Languages:** Python, C/C++, MATLAB
- **Tools:** PyTorch, Git, LaTeX, Markdown